§ 13.9 Change of variables

<u>1 variable case</u> $\int_a^b f(g(x))\, g'(x)\, dx = \int_{g(a)}^{g(b)} f(u)\, du$

eg $\int_1^2 x e^{-x^2} dx$ set $-x^2 = u$ $du = -2x\, dx$

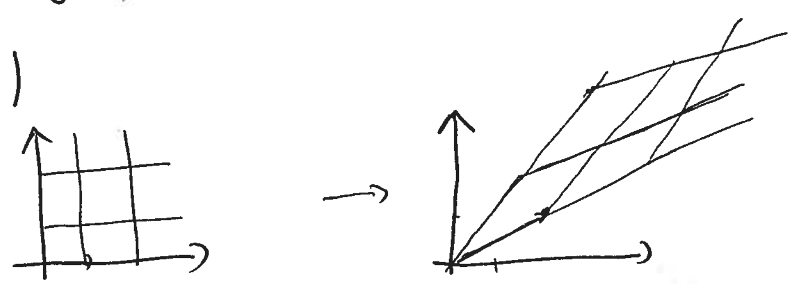
$= \int_{-1}^{-4} -\frac{1}{2} e^u\, du$

<u>2 variables</u> $x = x(u,v)$
$y = y(u,v)$

$G(u,v) = (x(u,v), y(u,v))$

$G: R^2 \longrightarrow R^2$ <u>1-1 map</u>

eg $G(u,v) = (2u+v,\ u+2v)$



$$\iint_R f(x,y)\, dx\, dy = \iint_S f(x(u,v), y(u,v))\, |J(u,v)|\, du\, dv$$

$G: \mathbb{R}^2 \longrightarrow \mathbb{R}^2$

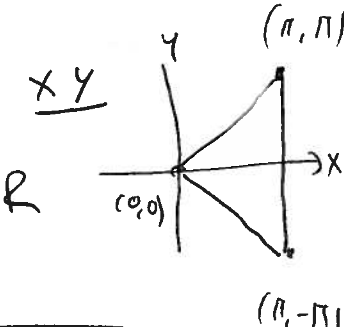
$R \longrightarrow S$

x-y plane u-v plane

$$J(u,v) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}$$

Ex $\iint_R \cos(x-y)\sin(x+y)\,dA$ $\quad$ R = triangle with vertices $(0,0)$ $(\pi,-\pi)$ (π,π)

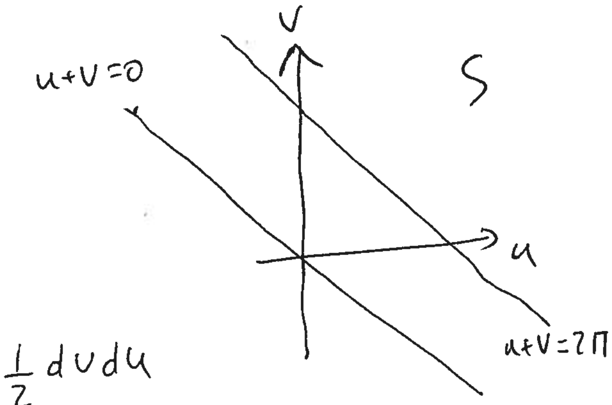
$u = x - y \quad v = x + y \iff x = \frac{1}{2}(u+v)$, $\; y = \frac{1}{2}(v-u)$



$R = \{(x,y) \mid 0 \le x \le \pi,\ -x \le y \le x\}$

So $\begin{cases} 0 \le \frac{u+v}{2} \le \pi \\ -\frac{1}{2}(u+v) \le \frac{1}{2}(v-u) \le \frac{1}{2}(u+v) \end{cases} \Rightarrow \begin{cases} 0 \le u+v \le 2\pi \\ -\frac{v}{2} \le \frac{v}{2} \ \& \ -\frac{u}{2} \le \frac{u}{2} \end{cases}$

or $0 \le v$ & $0 \le u$



$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{vmatrix} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$\iint_R \cos(x-y)\sin(x+y)\,dA = \iint_S \cos u \sin v \cdot \frac{1}{2}\,dv\,du$$

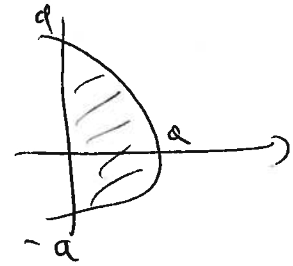
$$= \int_0^{2\pi}\int_0^{2\pi-u} \frac{\cos u \sin v}{2}\,dv\,du$$

$$= \frac{1}{2}\int_0^{2\pi} -\frac{\cos u \cos v}{2}\Big|_0^{2\pi-u} du = \frac{1}{2}\int_0^{2\pi} -\cos u \cos(2\pi-u) + \cos u \,du$$

$$= \frac{1}{2}\int_0^{2\pi} \cos u - \cos^2 u\,du = \frac{1}{2}\int_0^{2\pi} \cos u - \frac{1+\cos 2u}{2}\,du = \frac{1}{2}\left[\sin u - \frac{u}{2} - \frac{\sin 2u}{4}\right]_0^{2\pi}$$

$$= \frac{1}{2}\left[0 - \frac{2\pi}{2} - 0 - 0 + 0 + 0\right] = -\pi/2$$

Eg $\iint_R x\,dA$ $\qquad R = x^2+y^2 \le a^2 \qquad x \ge 0$



$$\int_{-a}^{a}\int_{0}^{\sqrt{a^2-y^2}} x\,dx\,dy = \qquad \begin{matrix} x = r\cos\theta \\ y = r\sin\theta \end{matrix} \rightsquigarrow \begin{matrix} r = \sqrt{x^2+y^2} \\ \theta = \tan^{-1}(y/x) \end{matrix}$$

$$S = \{ (r,\theta) \mid 0 \le r \le a \quad -\pi/2 \le \theta \le \pi/2 \}$$

$$J = \left| \begin{vmatrix} \partial x/\partial r & \frac{\partial x}{\partial \theta} \\ \partial y/\partial r & \partial y/\partial \theta \end{vmatrix} \right| = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = r$$

$$\iint_R x\,dA = \int_{-\pi/2}^{\pi/2}\int_{0}^{a} r\cos\theta \; r\,dr\,d\theta = \int_{-\pi/2}^{\pi/2} \frac{r^3}{3}\cos\theta \Big|_0^a d\theta = \frac{a^3}{3}\sin\theta \Big|_{-\pi/2}^{\pi/2}$$

$$= \frac{a^3}{3}\cdot 2$$

Eg spherical coordinates.

$$x = \rho\cos\theta\sin\phi$$
$$y = \rho\sin\theta\sin\phi$$
$$z = \rho\cos\phi$$

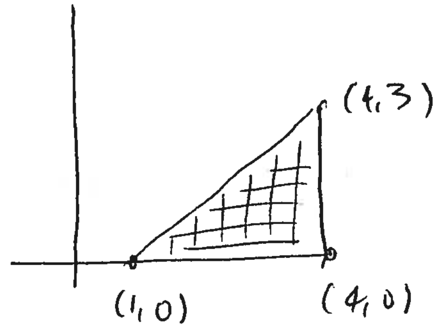
$$\begin{vmatrix} x_\rho & x_\theta & x_\phi \\ y_\rho & y_\theta & y_\phi \\ z_\rho & z_\theta & z_\phi \end{vmatrix} = \begin{vmatrix} \cos\theta\sin\phi & -\rho\sin\theta\sin\phi & \rho\cos\theta\cos\phi \\ \sin\theta\sin\phi & \rho\cos\theta\sin\phi & \rho\sin\theta\cos\phi \\ \cos\phi & 0 & -\rho\sin\phi \end{vmatrix}$$

$$= \cos\phi \left| -\rho^2\sin^2\theta\sin\phi\cos\phi - \rho^2\cos^2\theta\sin\phi\cos\phi \right) - \rho\sin\phi \left(\rho\cos^2\theta\sin^2\phi + \rho\sin^2\theta\sin^2\phi \right|$$

$$= -\cos\phi\,\rho^2\sin\phi\cos\phi - \rho\sin\phi\,\rho\sin^2\phi = -\rho^2\sin\phi(\cos^2\phi + \sin^2\phi)$$
$$= -\rho^2\sin\phi$$

$$|J(\rho,\theta,\phi)| = \rho^2\sin\phi$$

Ex $\iint_R \sqrt{\frac{x+y}{x-y}}\, dA$ R =



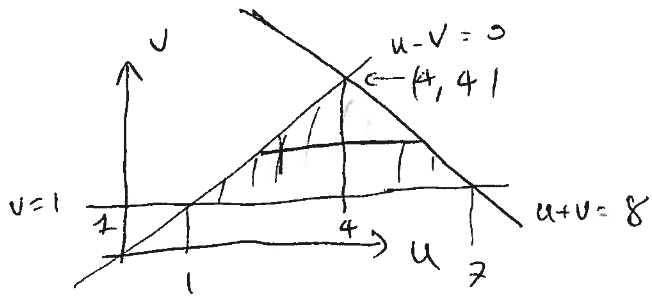
$u = x+y$ $\qquad x = \frac{u+v}{2}$

$v = x-y$ $\qquad y = \frac{u-v}{2}$

$J = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = \begin{vmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{vmatrix} = -\frac{1}{4} - \frac{1}{4} = -\frac{1}{2}$

$y \geq 0, \quad x \leq 4 \quad x-y \geq 1$

$u-v \geq 0 \quad u+v \leq 8 \quad v \geq 1$



$$\int_1^{\cancel{4}} \int_v^{8-v} \sqrt{\frac{u}{v}}\, \frac{1}{2}\, du\, dv$$

$$= \int_1^4 \frac{1}{\sqrt{v}} \frac{2}{\cancel{6}} u^{3/2} \Big|_v^{8-v} dv = \frac{1}{3}\int_1^4 \left(\frac{(8-v)^{3/2}}{\sqrt{v}} - v \right) dv \quad \text{.... hard}$$

$$= \int_1^4 \int_1^u \sqrt{\frac{u}{v}} \cdot \frac{1}{2}\, dv\, du + \int_4^7 \int_1^{8-u} \frac{1}{2}\sqrt{\frac{u}{v}}\, dv\, du =$$

$$= \int_1^4 \frac{2}{2}\sqrt{uv}\Big|_1^u du + \int_4^7 \frac{2}{2}\sqrt{uv}\Big|_1^{8-u} du = \frac{1}{2}\int_1^4 2u - 2\sqrt{u}\, du + \int_4^7 2\sqrt{u(8-u)} - 2\sqrt{u}\, du$$

$$= \frac{u^2}{2}\Big|_1^4 - \frac{4}{\cancel{6}} u^{3/2}\Big|_1^7 + \cancel{2}\int_4^7 \sqrt{-(u-4)^2 + 16}\, du = \frac{16-1}{2} - \frac{\cancel{4}}{3}\left(7^{3/2} - 1\right) + \cancel{2}\int_4^7 4\sqrt{1 - \left(\frac{u}{4} - 1\right)^2}\, du$$

$$= \frac{15}{2} - \frac{\cancel{4}}{3}\left(7^{3/2} - 1\right) + \cancel{4}\left[-4\cos^{-1}\left(\frac{u}{4} - 1\right)\right]_4^7 = \dots \qquad D_x \cos^{-1} x = \frac{-1}{\sqrt{1-x^2}}$$

Ex Find the volume and center of mass for the ellipsoid $(x/a)^2 + (y/b)^2 + (z/c)^2 = 1$ via the change of variable $x = ua$, $y = vb$, $z = wc$

Ex Compute the following when R is the triangle (1,0), (4,0), (4,3)

$\iint_R \ln\left(\frac{x+y}{x-y}\right) dA$, $\iint_R \sin(\pi(2x-y))\cos(\pi(y-2x))\, dA$, $\iint_R (2x-y)\cos(y-2x)\, dA$